Azure Machine Learning

scikit-learn

권기웅

1. 실행

1.1 kNN 실행

```python
import numpy as np
from sklearn.neighbors import KNeighborsClassifier

feature = np.array([[80, 20], [95, 5], [10, 90], [90, 10], [5, 95]])
label = np.array(["액션", "액션", "조폭", "액션", "조폭"])

knc = KNeighborsClassifier(3)
knc.fit(feature, label)

fight = int(input("싸움 : "))
yok = int(input("욕 : "))
predictData = np.array([[fight, yok]])
result = knc.predict(predictData)
print(result[0])
```

[1] ✓ 6.7s

```
액션
```

2. 활용

2.1 모델 등록

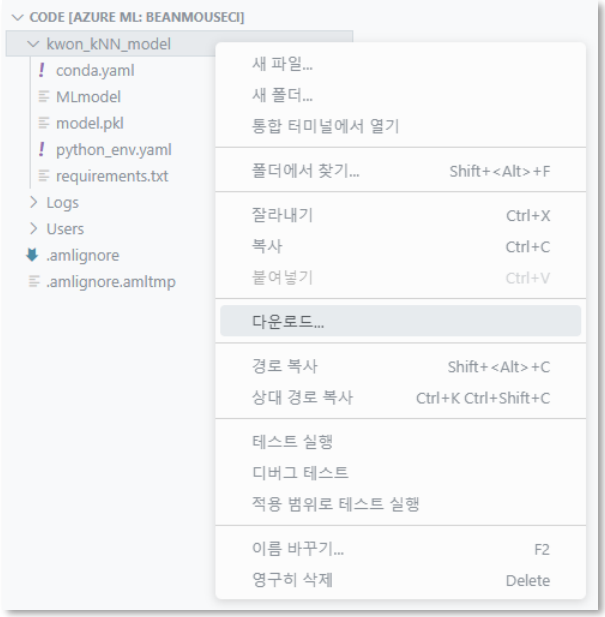
2.1.1 모델 만들기

```python
import numpy as np
from sklearn.neighbors import KNeighborsClassifier
import mlflow

mlflow.start_run()

feature = np.array([[80, 20], [95, 5], [10, 90], [90, 10], [5, 95]])
label = np.array(["액션", "액션", "조폭", "액션", "조폭"])

knc = KNeighborsClassifier(3)
knc.fit(feature, label)

fight = 30
yok = 70
predictData = np.array([[fight, yok]])
result = knc.predict(predictData)
print(result[0])

mlflow.sklearn.save_model(
    sk_model=knc,
    path="kwon_kNN_model",
)

mlflow.end_run()
```

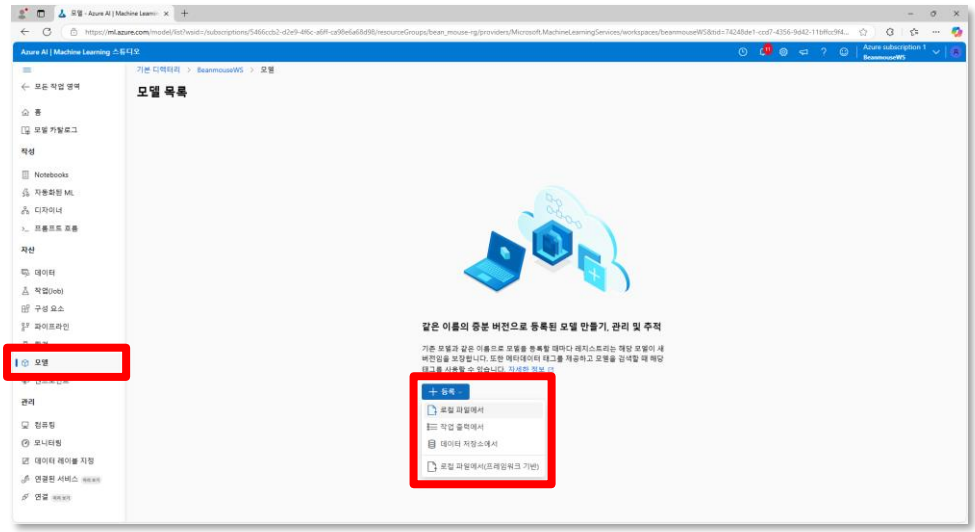
[2] ✓ 6.8s

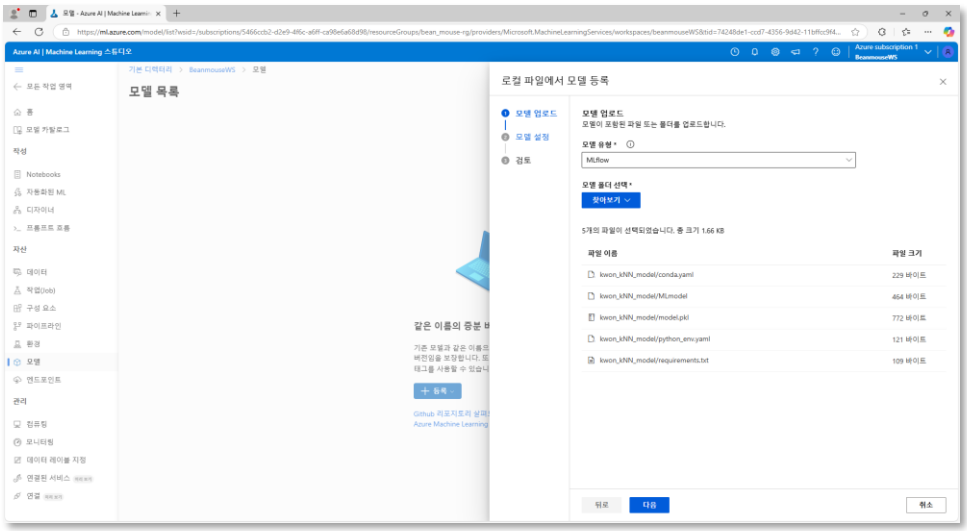
```
/anaconda/envs/azureml_py310_sdkv2/lib/python3.10/site-packages/azureml/
  from google.protobuf import service as _service
조폭
2025/02/05 09:54:38 INFO mlflow.tracking._tracking_service.client: 🏃 View
2025/02/05 09:54:38 INFO mlflow.tracking._tracking_service.client: 🧪 View
```

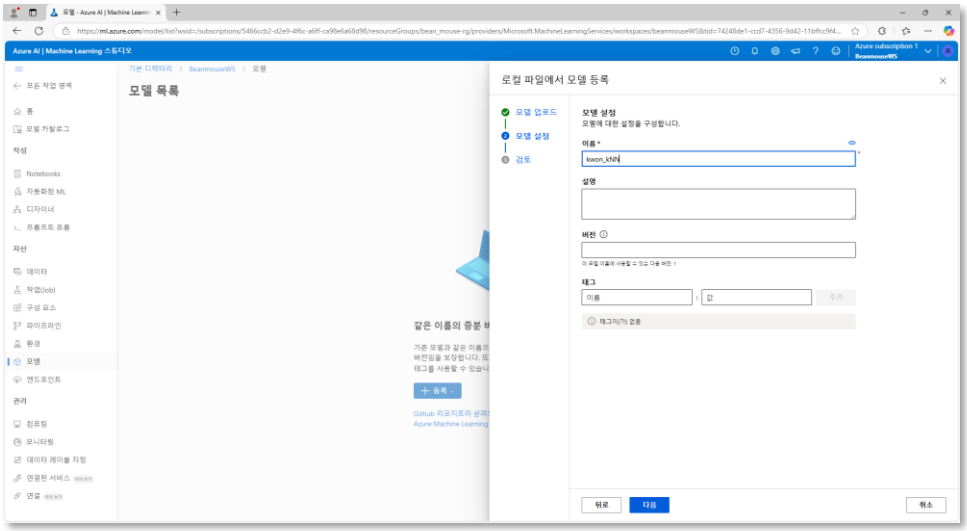
2.1.2 다운로드



2.1.3 모델 등록




Azure AI | Machine Learning 스튜디오
모델 목록
로컬 파일에서 모델 등록
모델 업로드
모델 설정
검토
모델 업로드
모델이 포함된 파일 또는 폴더를 업로드합니다.
모델 유형 *
MLflow
모델 폴더 선택 *
찾아보기
5개의 파일이 선택되었습니다. 총 크기 1.66 KB
파일 이름
파일 크기
kwon_kNN_model/conda.yaml
229 바이트
kwon_kNN_model/MLmodel
464 바이트
kwon_kNN_model/model.pkl
772 바이트
kwon_kNN_model/python_env.yaml
121 바이트
kwon_kNN_model/requirements.txt
109 바이트
뒤로
다음
취소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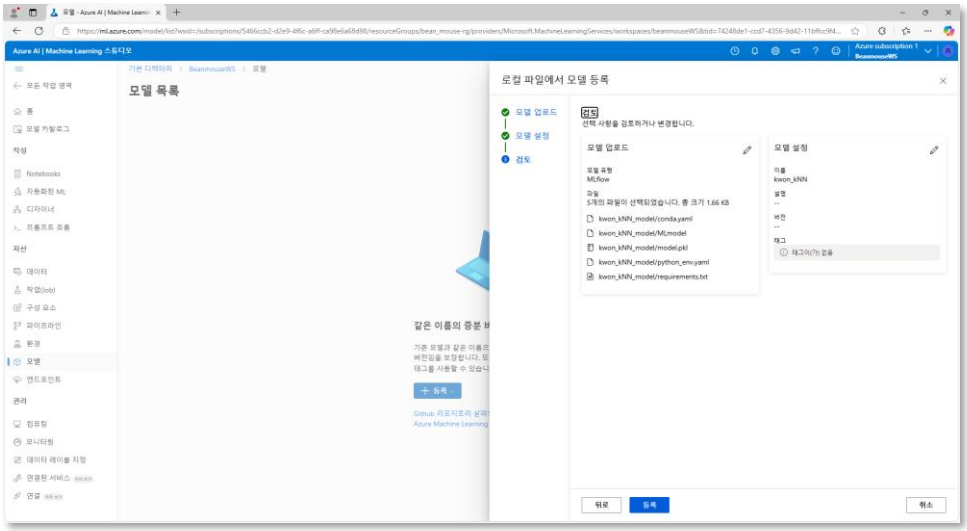
모델 업로드
모델 설정
검토
모델 설정
모델에 대한 설정을 구성합니다.
이름 *
kwon_kNN
설명
버전
태그
이름
값
추가
태그이(가) 없음
뒤로
다음
취소


Azure AI | Machine Learning 스튜디오
모델 목록
로컬 파일에서 모델 등록
모델 업로드
모델 설정
검토
검토
선택 사항을 검토하거나 변경합니다.
모델 업로드
모델 유형
MLflow
파일
5개의 파일이 선택되었습니다. 총 크기 1.66 KB
kwon_kNN_model/conda.yaml
kwon_kNN_model/MLmodel
kwon_kNN_model/model.pkl
kwon_kNN_model/python_env.yaml
kwon_kNN_model/requirements.txt
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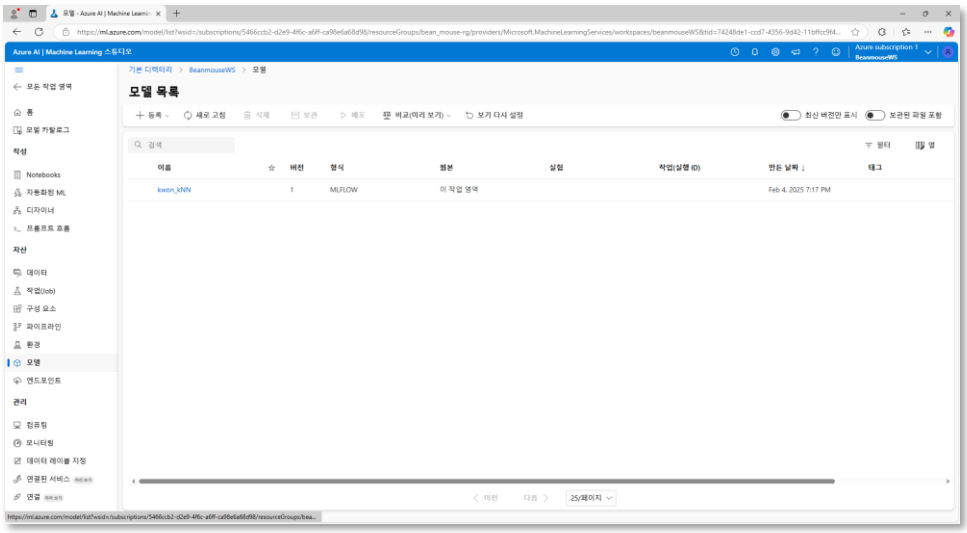
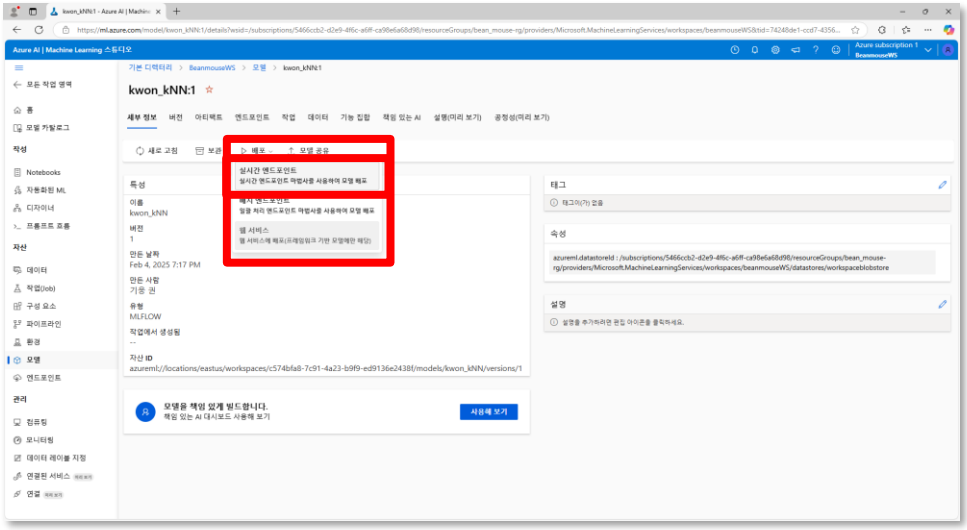
kwon_kNN
설명
버전
태그
태그이(가) 없음
뒤로
등록
취소

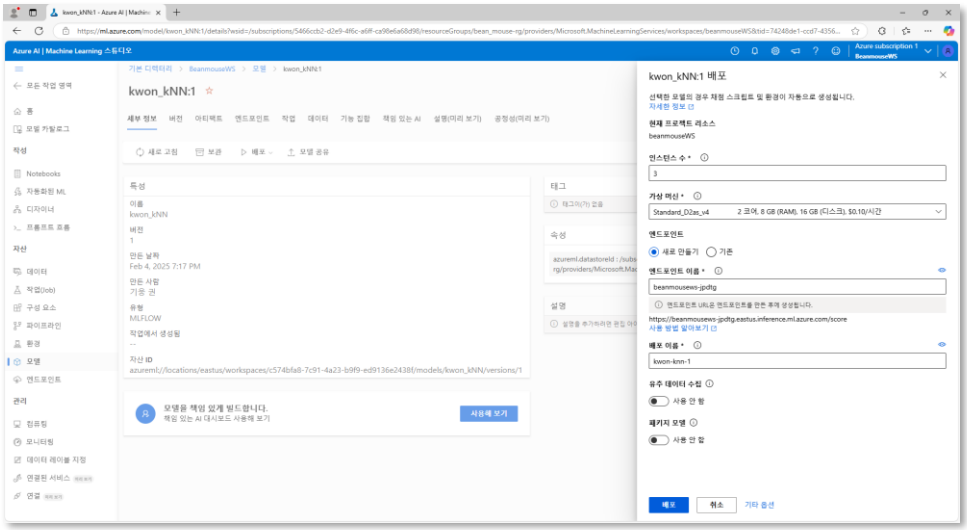
2.2 엔드포인트 만들기

2.2.1 만들어진 모델 들어가서



2.2.2 엔드포인트 만들기





한 세월

2.3 클라이언트 프로그램

```python
import urllib.request
import json
import os
import ssl


def allowSelfSignedHttps(allowed):
    if (
        allowed
        and not os.environ.get("PYTHONHTTPSVERIFY", "")
        and getattr(ssl, "_create_unverified_context", None)
    ):
        ssl._create_default_https_context = ssl._create_unverified_context


allowSelfSignedHttps(True)

data = {"input_data": {"columns": ["fight", "yok"], "index": [0], "data": [[80, 50]]}}


body = str.encode(json.dumps(data))

url = "https://beanmousews-jpdtg.eastus.inference.ml.azure.com/score"

api_key = "sPXg6DLNUyTv4InSn47VvAPLUgCuaRzz"
if not api_key:
    raise Exception("A key should be provided to invoke the endpoint")


headers = {"Content-Type": "application/json", "Authorization": ("Bearer " + api_key)}

req = urllib.request.Request(url, body, headers)

try:
    response = urllib.request.urlopen(req)

    result = response.read()
    result = json.loads(result)[0]
    print(result)
except urllib.error.HTTPError as error:
    print("The request failed with status code: " + str(error.code))

    print(error.info())
    print(error.read().decode("utf8", "ignore"))
```

3. DB 연동

3.1 OracleDB

3.1.1 라이브러리 설치

```
%pip install oracledb
```

```
%pip install oracledb
[4] ✓ 1.3s
Collecting oracledb
  Using cached oracledb-2.5.1-cp310-cp310-manylinux_2_17_x86_64.m
Requirement already satisfied: cryptography>=3.2.1 in /anaconda/e
Requirement already satisfied: cffi>=1.12 in /anaconda/envs/azure
Requirement already satisfied: pycparser in /anaconda/envs/azurem
Using cached oracledb-2.5.1-cp310-cp310-manylinux_2_17_x86_64.man
Installing collected packages: oracledb
Successfully installed oracledb-2.5.1
Note: you may need to restart the kernel to use updated packages.
```

3.1.2 프로그램

```python
import pandas as pd
import numpy as np
from oracledb import connect
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import MinMaxScaler
import mlflow

mlflow.start_run()

mms = MinMaxScaler()
knc = KNeighborsClassifier(3)

con = connect("kwon/1@sd-beanmouse.duckdns.org:15211/xe")
sql = "select * from feb04_movie"
df = pd.read_sql(sql, con)
con.close()
print(df)

feature = df[["M_FIGHT", "M_YOK"]].to_numpy()
label = df["M_GENRE"].to_numpy()
feature = mms.fit_transform(feature)
knc.fit(feature, label)

fight = 30
yok = 70
predictData = np.array([[fight, yok]])
predictData = mms.transform(predictData)
result = knc.predict(predictData)[0]
print(result)

mlflow.sklearn.save_model(
    sk_model=knc,
    path="kwon_kNN_model2",
)

mlflow.end_run()
```

[22]

```
   M_FIGHT  M_YOK M_GENRE
0       80     20     액션
1       95      5     액션
2       10     90     조폭
```

3.2 MongoDB

3.2.1 라이브러리 설치

```
%pip install pymongo
```

```
[14]
%pip install pymongo

Collecting pymongo
  Downloading pymongo-4.11-cp310-cp310-manylinux_2_17_x86_64.many
Collecting dnspython<3.0.0,>=1.16.0 (from pymongo)
  Downloading dnspython-2.7.0-py3-none-any.whl.metadata (5.8 kB)
Downloading pymongo-4.11-cp310-cp310-manylinux_2_17_x86_64.manyli
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 36.8 MB/s eta 0:0
Downloading dnspython-2.7.0-py3-none-any.whl (313 kB)
Installing collected packages: dnspython, pymongo
Successfully installed dnspython-2.7.0 pymongo-4.11
Note: you may need to restart the kernel to use updated packages.
```

3.2.2 프로그램

```
[24]
from pymongo import MongoClient
from sklearn.naive_bayes import MultinomialNB
from sklearn.feature_extraction.text import CountVectorizer
import mlflow

mlflow.start_run()

con = MongoClient("sd-beanmouse.duckdns.org:2710")
db = con.xe
data = []
label = []
for b in db.naverBlog.find():
    data.append(b["title"] + " " + b["desc"])
    label.append(b["label"])
con.close()

cv = CountVectorizer()
cvResult = cv.fit_transform(data).toarray()
mnb = MultinomialNB()
mnb.fit(cvResult, label)

sentence = "오늘은 맑음"
sentenceCvResult = cv.transform([sentence]).toarray()
result = mnb.predict(sentenceCvResult)[0]
print(result)

mlflow.sklearn.save_model(
    sk_model=mnb,
    path="kwon_mnb_model",
)

mlflow.end_run()
```

```
광고
2025/02/05 09:37:12 INFO mlflow.tracking._tracking
2025/02/05 09:37:12 INFO mlflow.tracking._tracking
```